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BEST PRACTICE SERIES

Piston, Rod, and Liner Staging Equipment

Application | Maintenance | Site Management | **Component Rebuild** | Safety | MARC Management

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1.0 Introduction

Effective staging equipment for pistons, rods, & liners (PR&L) is a cost effective way to quickly improve engine repair/rebuild operations. Parts staging is typically a non-value adding step in the rebuild process and this Best Practice reduces staging associated costs and technician efforts at several task levels.

Special staging cabinets allows product to “pass through” them with little effort. They provide an efficient way to stage PR&L after they’ve been rebuilt. They also protect the components from contamination and handling damage, and automatically transfer the component out of the rebuild bay. Well-protected components incur less incidental damage and provide higher quality to the customer.

In summary the enhanced staging methods/equipment:

- Allows technician to spend minimal effort in staging rebuilt PR&L.
- Provides contamination and handling damage protection to the product.
- Automatically moves the rebuilt component from inside the rebuild area to outside the area.

2.0 Best Practice Description

PR&L are often delivered to the rebuild area on carts or other transport device. These carts, and other miscellaneous baskets area typically not very space efficient. Once rebuilt, the components should stay at rebuild area staged until needed. When the assembly job calls for the components, the pieces are loaded and transported to the necessary assembly bay. Incoming and outgoing staging consumes excessive floor space, and uses lots of resources to get protected product to the assembly bay.

Enhanced piston, rod, and liner staging equipment has several features:

- The cabinet provides high-density component staging. PR&L are stored at several levels in the cabinets, providing storage for several jobs of components.
- Liners are protected by in tracks they can be rolled through. Assembled pistons and rods are hug in multi-level racks, which saddle the pistons and allow the rod to hang from the piston pin. These tracks and racks protect the parts from storage damage.
- The cabinets are enclosed with Plexiglas doors, allowing for job parts identification, and protecting against contamination.
- The cabinet doors are on opposite sides of the staging cabinet. This allows the piston, rod, and liner rebuild technician to transfer rebuilt parts out of the working area, into the transport aisle, by simply loading the cabinet through the work bay cabinet door.
- If the PR&L rebuild bay is located next to the short-block assembly bay, the parts would automatically be delivered to the general assembly activity. The assembly technician would just pull the current job’s PR&L from behind the assembly bay cabinet door.

The staging cabinets:

- Are sized for the dealer's common product.
- Have smooth painted interior/exterior designed with few areas to catch dust/dirt/crime for fast/easy cleaning and professional image.
- Use Plexiglas (clear) doors to identify what product is rebuilt.
- For liners, include rolling tracks that are sloped towards the outgoing door.

These cabinets provide the most efficiency if located between the PR&L rebuild area and the general short block or general engine assembly bays. This placement eliminates one stage of handling and greatly reduces incidental handling damage exposure.

3.0 Implementation Steps

1. Determine area in shop to be designated for PR&L rebuilding.
    a. Review the rebuild area in relation to downstream processes, such as PR&L installation.
    b. Locating the rebuild area next to the bay where they are installed can eliminate one parts move.
    c. Establish PR&L cabinet location within the PR&L rebuild area.

2. Determine most common PR&L sizes, and typical volumes in staging.
    a. Design staging cabinets to accommodate largest parts/components.
    b. Establish cabinet usability/finish to assure easy cleaning/maintenance and professional appearance.
    c. If PR&L rebuild area is not next to the installation bay(s):
    d. Establish equipment needed to transfer/stage PR&L in the installation bay(s).
    e. Plan for transfer/staging equipment storage between uses (close to rebuild area is better).

3. Develop/adjust and document PR&L rebuild procedure as needed.

4. Establish new/adjusted repair/rebuild time requirement targets as needed.

5. Present strategy to shop and other related employees.
    a. <u>Eliminate</u> the current processes/tooling/equipment strategy (one best way).
    b. Include inter-department use issues as needed.

6. Install the layout/equipment changes into the shop
   a. This may occur in phases, as needed, such as shop arrangement, then equipment final design/fabrication.
   b. Promote change feedback as installation may reveal new problems.

7. Train employees to use the new procedures/equipment. Follow-up/enforce immediately.

8. Establish equipment maintenance responsibility.

9. Review process/equipment performance once established.

10. Establish and install any adjustments as needed.

4.0 Benefits

Improved Efficiency

- Well designed/organized staging equipment reduces technician staging efforts setup and teardown time. Smooth and orderly product flow simplifies process and reduces administrative/rebuild labor.
- Product staged with like-product reduces space needed.
- Assembly delays are minimized with fewer misplaced/damaged PR&L. PRL are seldom replaced at the last minute due to unexpected handling damage. Technicians are not searching the shop for missing PR&L pieces.

Improved job flexibility

- PR&L are not used/delivered until needed. This makes it easier to change jobs in a bay without lots of staged parts to move.

Increases Quality

- Effective staging equipment reduces handling related damage, increasing component service life, and promoting professionalism with technicians.
- Parts incur less contamination and handling damage, providing more component service life.

5.0 Resources Required

Facility

- Re-organize workstation to allow for basin placement, incoming and outgoing staging, and a workbench as needed.

Support Equipment

- Well-built cabinet costs will vary since they are typically custom-built at the dealer.

People

- Establish change concerns with team
- Establish process, layout, and equipment improvements through team

Training

- Define/establish/enforce new processes, roles and responsibilities with production teams
- Remove replaced items (staging equipment) from operations to sustain change

6.0 Supporting Attachments / References

None

7.0 Related Best Practices

None

8.0 Acknowledgements

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